Name-Surname: .... ID: ....

01286326 THEORY OF COMPUTATION
ASSIGNMENT 1

PROBLEM 1

Suppose M_1 is a finite automaton $(Q, \Sigma, \delta, q_0, F)$ where

- $Q = \{q_0, q_1, q_2, q_3, q_4\}$
- $\Sigma = \{0,1\}$
- δ is described as

	0	1
q_0	q_1	q_3
q_1	q_2	q_1
q_2	q_4	q_1
q_3	q_3	q_3
q_4	q_4	q_2

- q_0 is the initial state,
- $F = \{q_1, q_4\}$

1.1) Draw a state diagram of machine M_1.

1.2) Write the computation path of M_1 on the string 010011. Is this string accepted by M_1?

1.3) Find all the strings of length no greater than 4 accepted by M_1.

PROBLEM 2

Let L_1 and L_2 be the following languages on the alphabet $\Sigma = \{0,1\}$:

$$L_1 = \{w \in \Sigma^* \mid w \text{ contains an even number of occurrences of } 1\ \}$$

$$L_2 = \{w \in \Sigma^* \mid w \text{ contains no adjacent occurrences of } 1\}.$$

2.1 Design (deterministic) finite automata M_1 and M_2 that recognize L_1 and L_2, respectively.

2.2 Construct a product automaton of M_1 and M_2 that recognizes the language $L_1 \cap L_2$.

2.3 Construct a product automaton of M_1 and M_2 that recognizes the language $L_1 \cup L_2$.

PROBLEM 3

3.1 Design a (deterministic) finite automaton that recognizes the following language L on the alphabet $\Sigma = \{0,1\}$:

$$L = \{\ 0011, 1010, 1110, 0100, 111, 01\ \}.$$

3.2 Design a (deterministic) finite automaton that recognizes the following language L' on the alphabet $\Sigma = \{0,1\}$:

$$L' = \{w \in \Sigma^* \mid w \text{ is a binary representation of a non-negative integer divisible by } 3\}.$$

PROBLEM 4

Suppose M_2 is a nondeterministic finite automaton $(Q, \Sigma, \delta, q_0, F)$ where

- $Q = \{p_0, p_1, p_2, p_3, p_4\}$
- $\Sigma = \{0,1\}$
- δ is described as

	ϵ	0	1
p_0	$\{p_1\}$	$\{p_4\}$	$\{\}$
p_1	$\{\}$	$\{p_1\}$	$\{p_1, p_2\}$
p_2	$\{\}$	$\{\}$	$\{p_3\}$
p_3	$\{\}$	$\{\}$	$\{\}$
p_4	$\{\}$	$\{p_4\}$	$\{\}$

- p_0 is the initial state,
- $F = \{p_3, p_4\}$

4.1) Draw a state diagram of machine M_2.

4.2) Write a computation tree of M_2 on the string 01011. Is this string accepted by M_2?

4.3) Apply subset construction to construct a (deterministic) finite automaton that is equivalent to M_2.

PROBLEM 5

Construct a nondeterministic finite automaton recognizing the following language $\Sigma^* - (L_1 L_2)$ where L_1 and L_2 are languages on the alphabet $\Sigma = \{0,1\}$ given as follows:

$$L_1 = \{w \in \Sigma^* \mid w \text{ contains an even number of occurrences of } 1\ \}$$

$$L_2 = \{w \in \Sigma^* \mid w \text{ contains no adjacent occurrences of } 1\}.$$